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(54) **MACHINE LEARNED FEATURE
RECOMMENDATION ENGINE IN A
DIGITAL TRANSACTION MANAGEMENT
PLATFORM**

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(57) **ABSTRACT**

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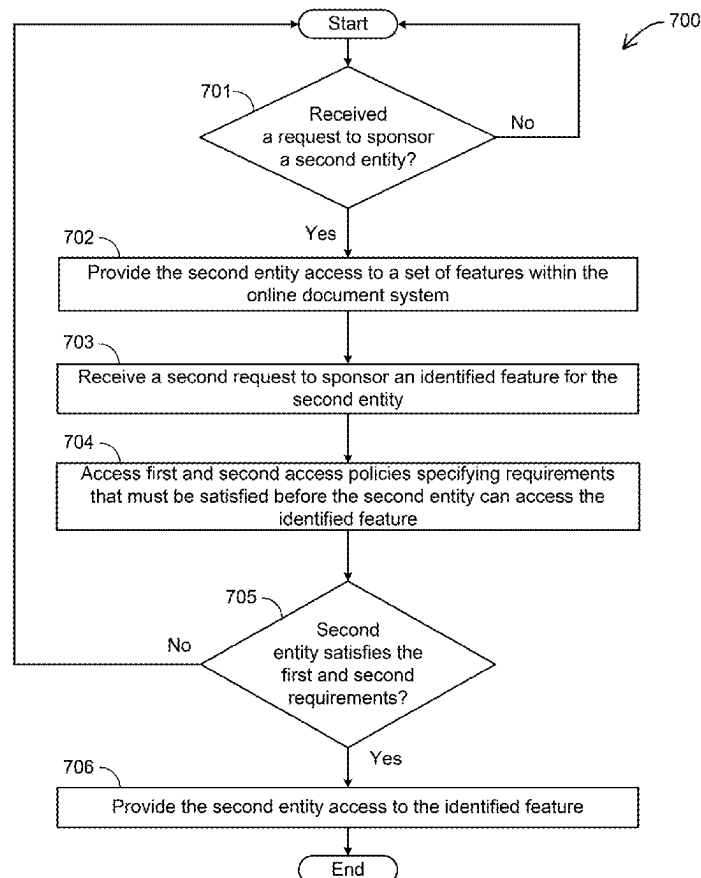
Related U.S. Application Data

(63) Continuation of application No. 18/512,382, filed on
Nov. 17, 2023, now Pat. No. 12,339,929, which is a
continuation of application No. 16/870,557, filed on
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An online document system provides a recommendation for one or more features within the online document system to an entity. The online document system accesses a set of feature training data to train a machine learning model. The set of feature training data may describe characteristics of entities associated with the online document system and historical activity associated with the entities' usage of the online document system's features. The machine learning model may be configured to identify a feature to recommend to an entity based on the entity's characteristics and history of using other features within the online document system. For example, data representing the entity's user accounts and use of an electronic signature feature is used by the machine learning model to identify a document authentication feature to recommend to the entity. The online document system may then provide the identified feature in a recommendation to the entity.



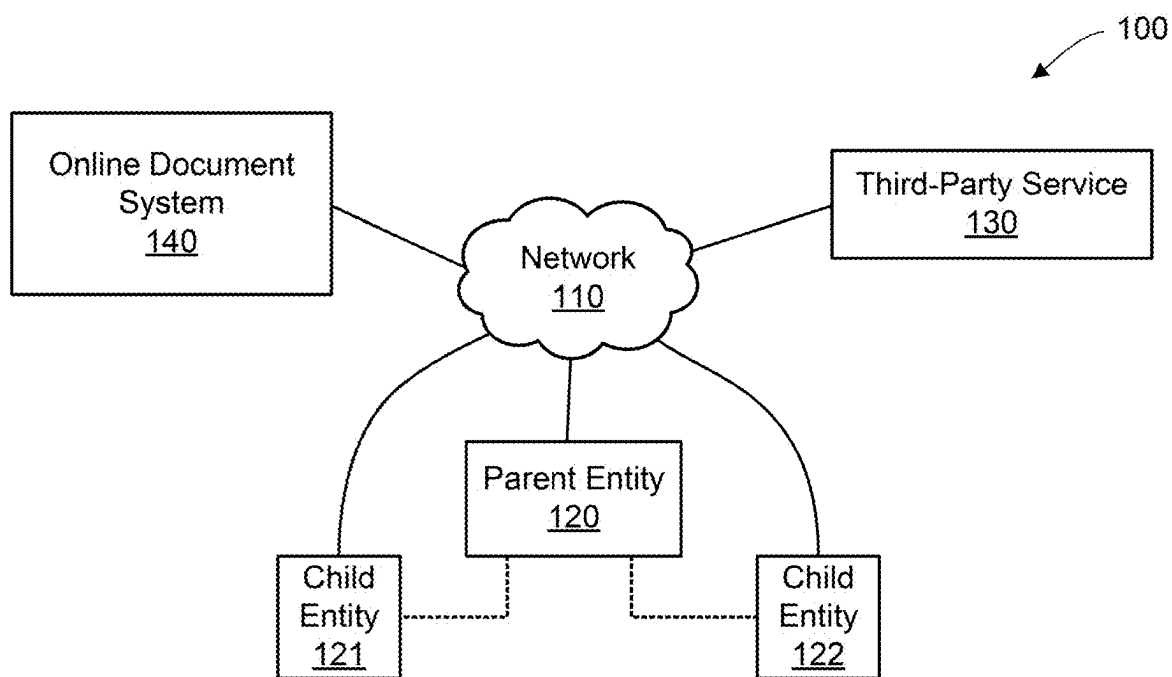


FIG. 1

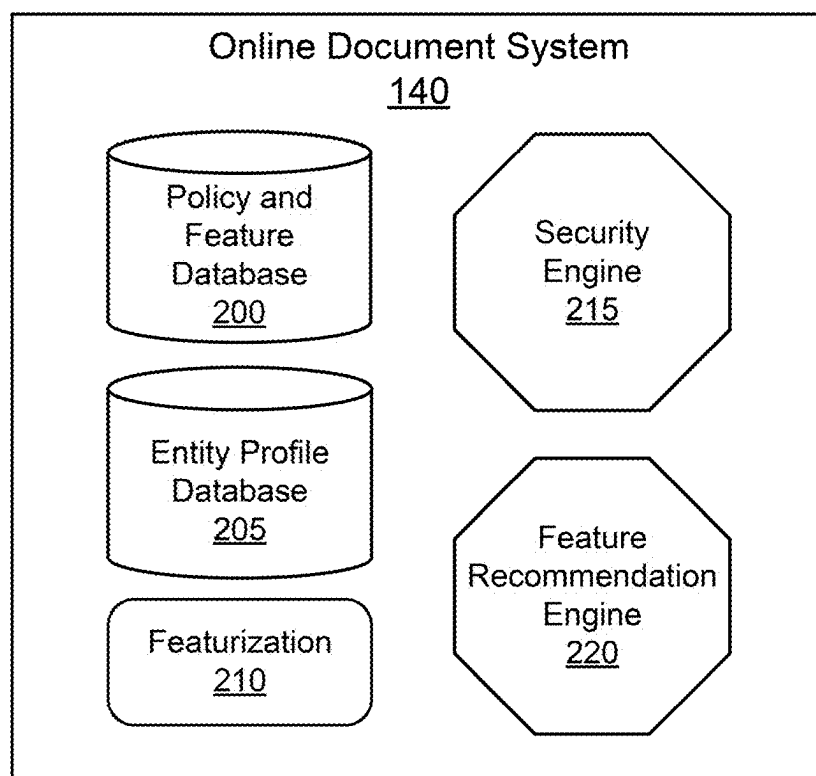


FIG. 2

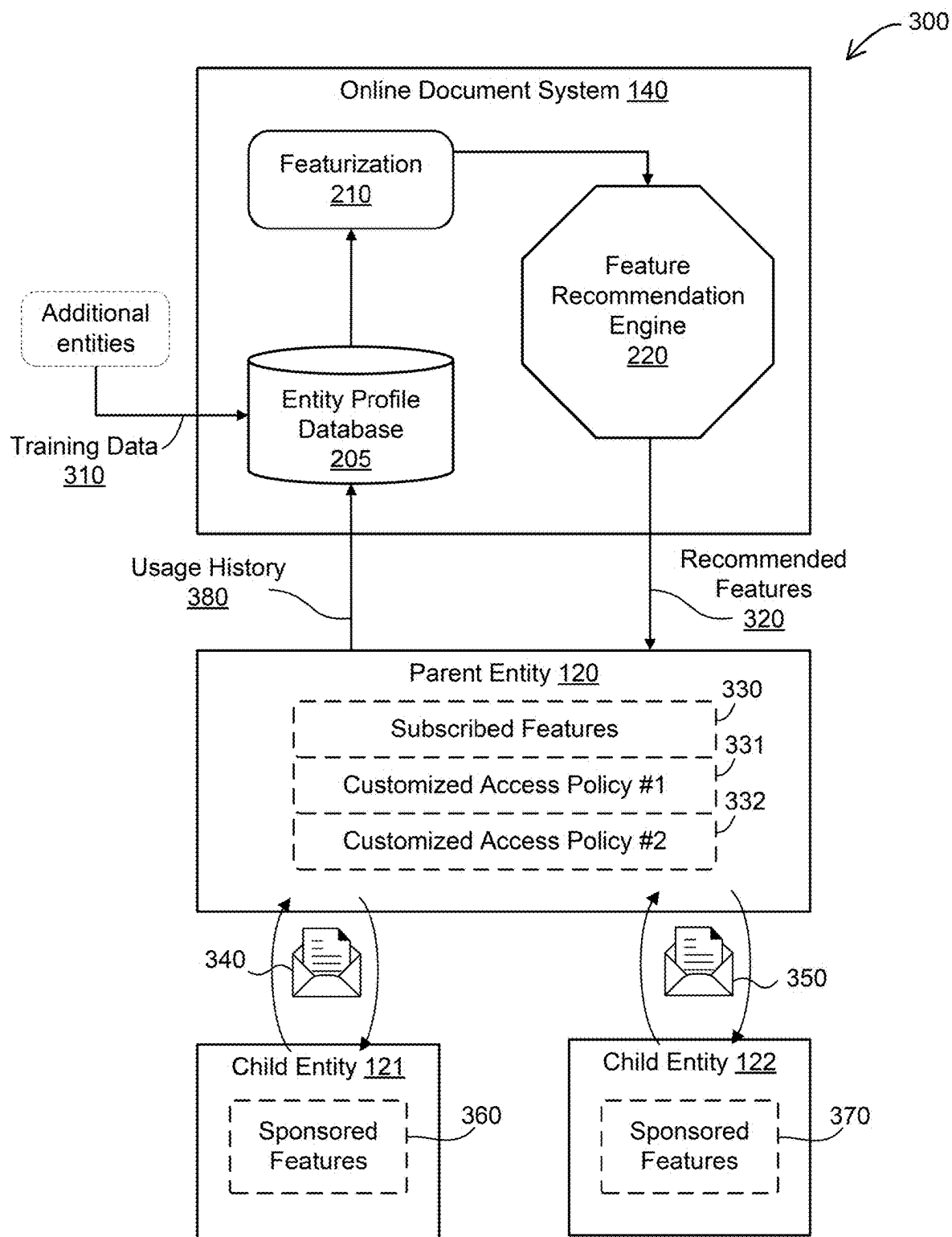


FIG. 3

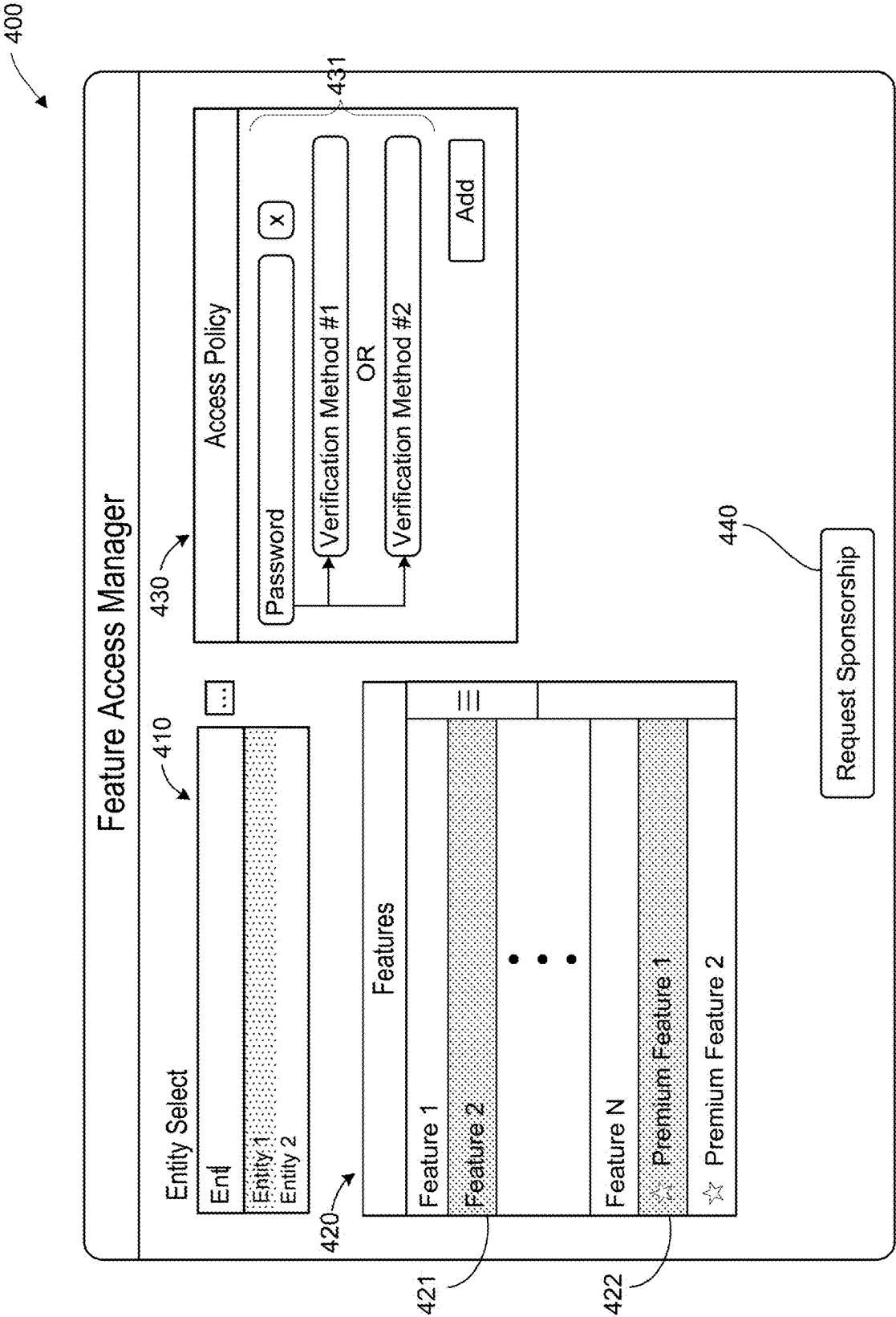
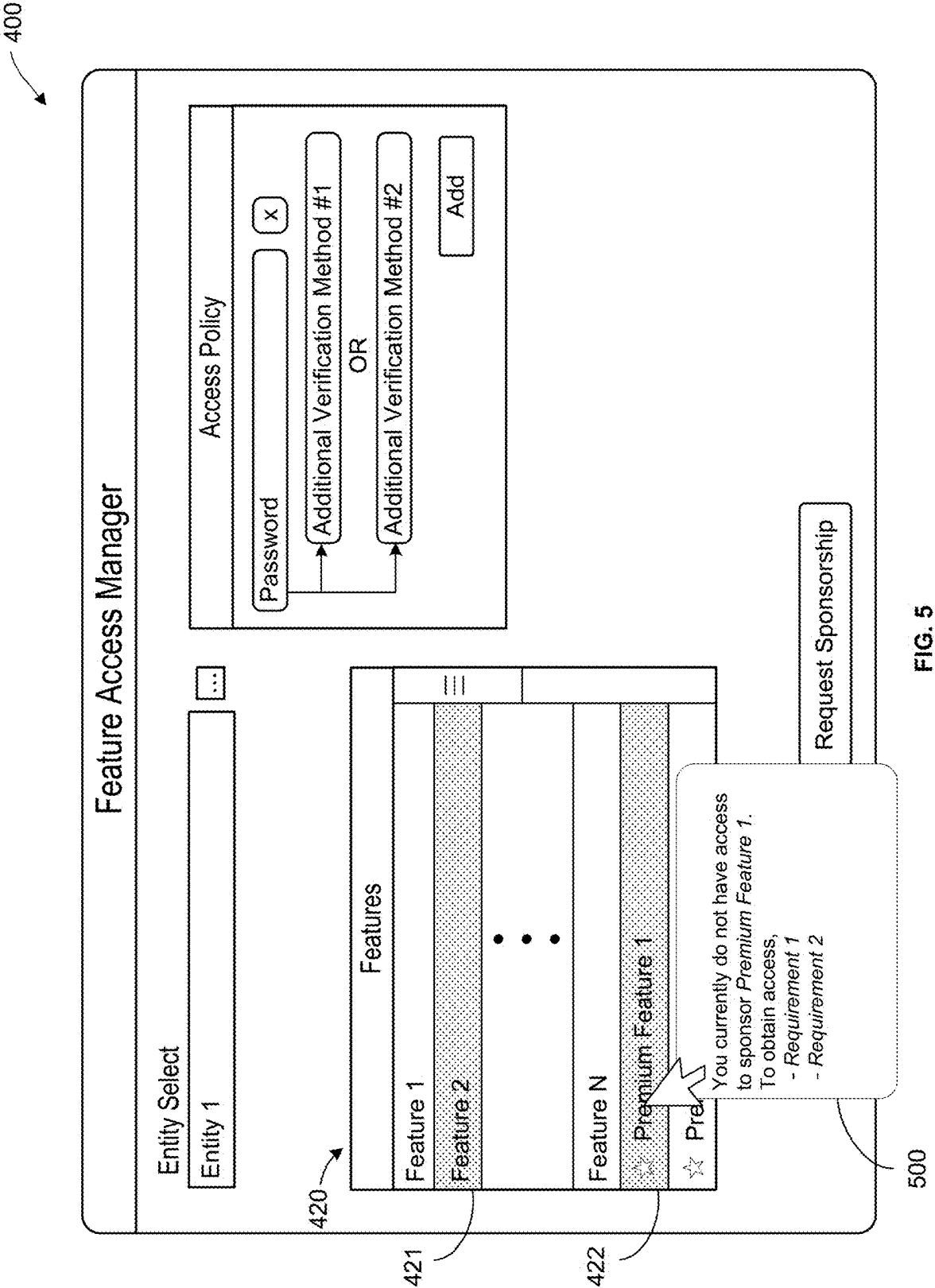


FIG. 4



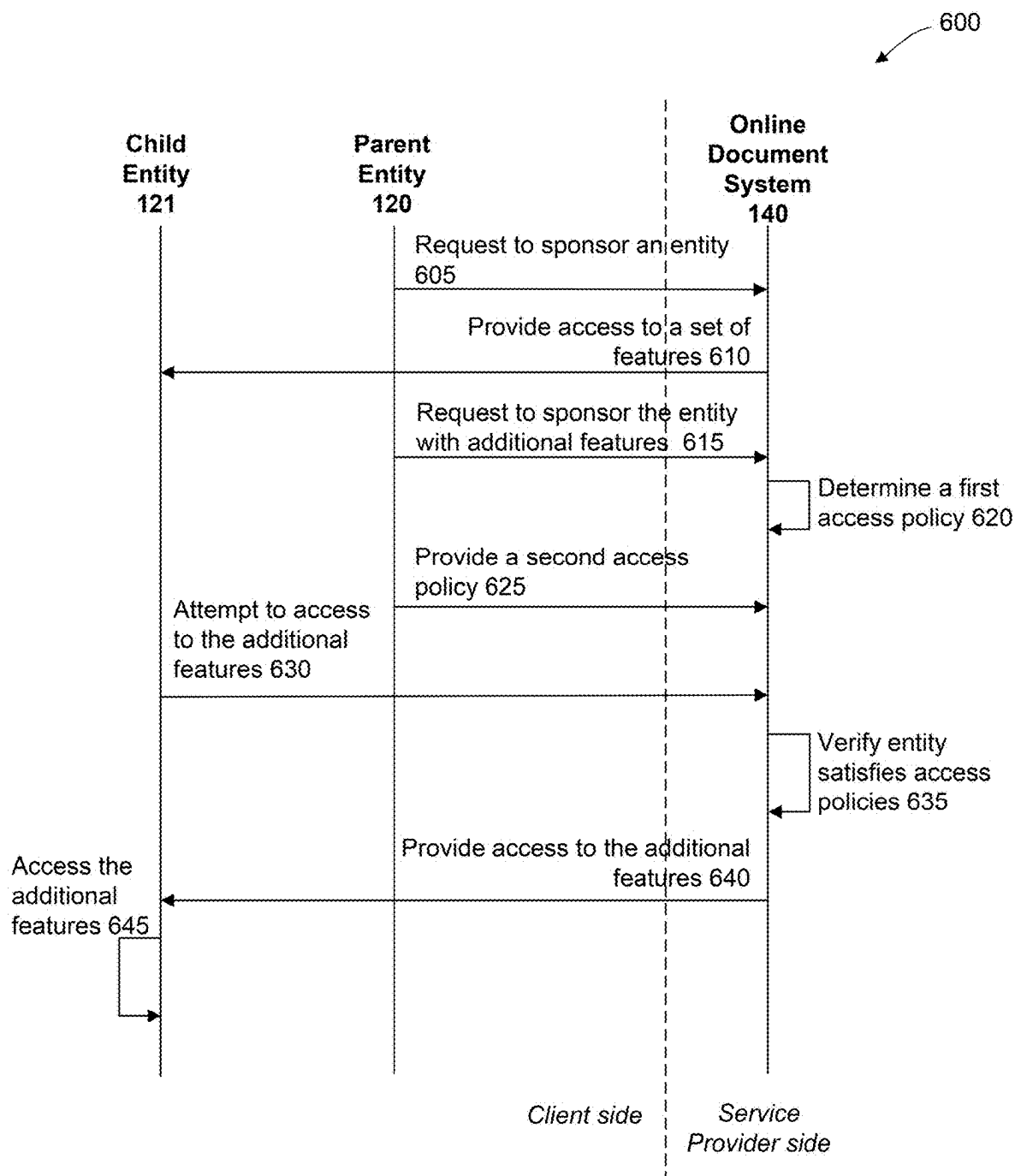
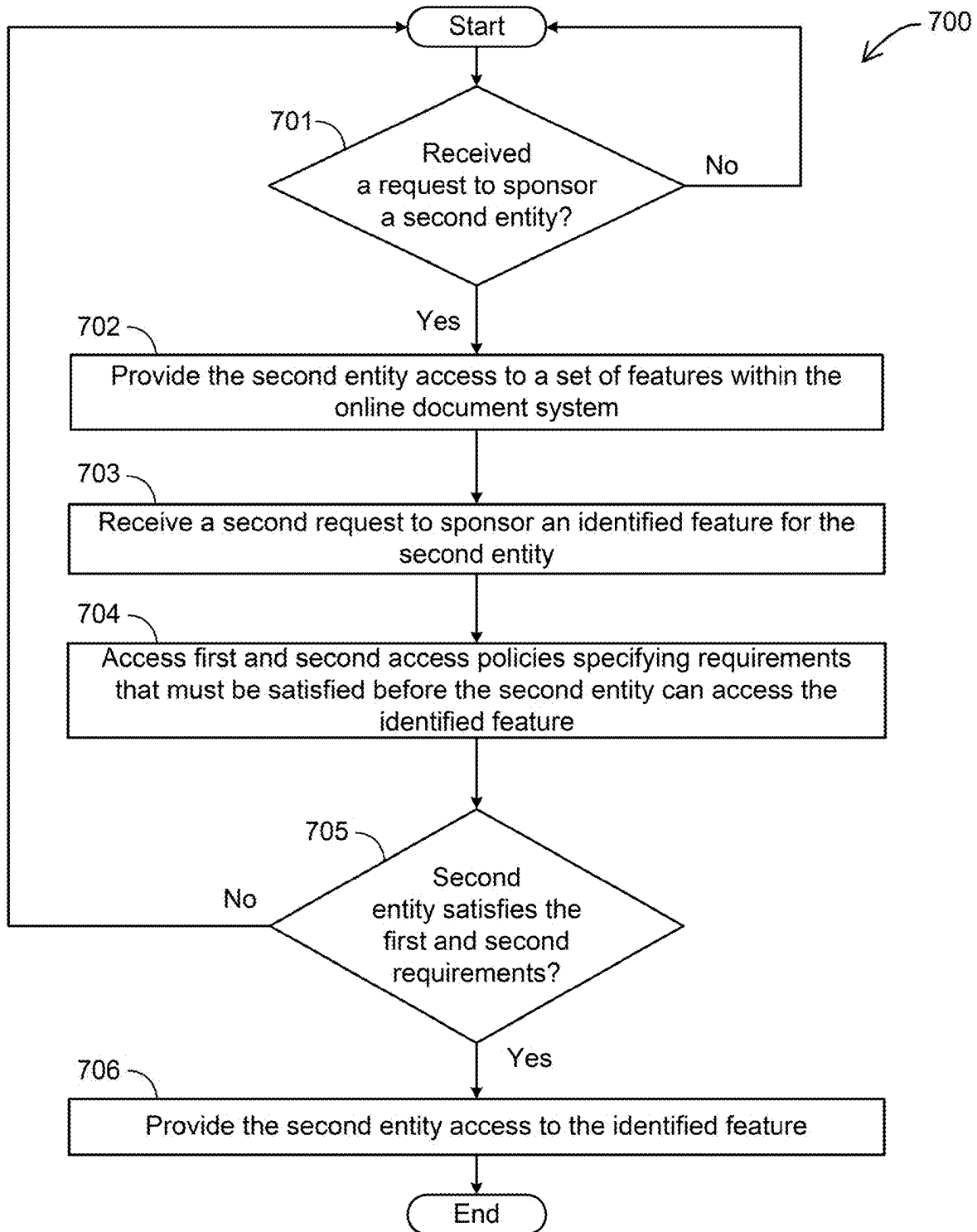


FIG. 6

**FIG. 7**

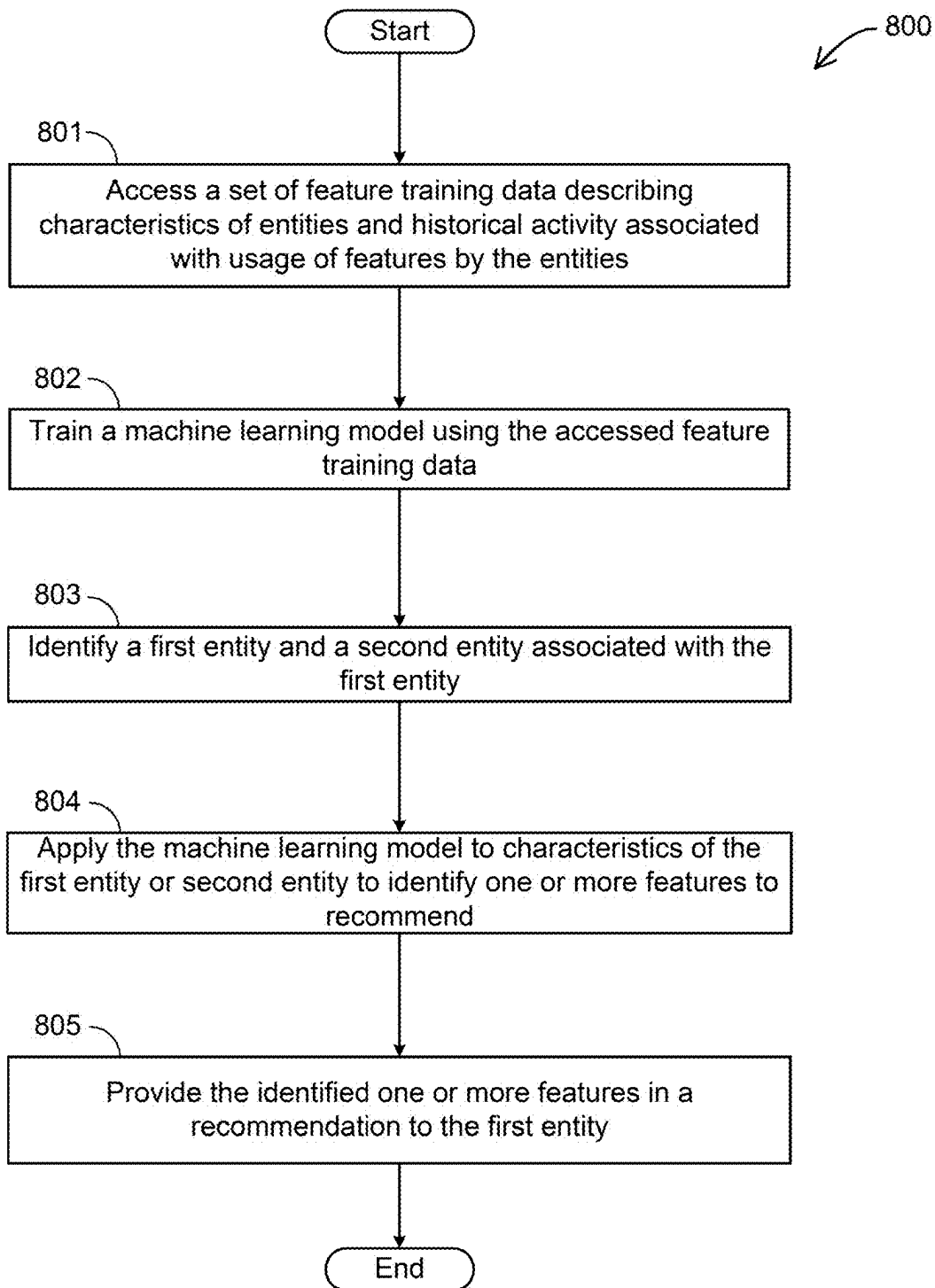


FIG. 8

**MACHINE LEARNED FEATURE
RECOMMENDATION ENGINE IN A
DIGITAL TRANSACTION MANAGEMENT
PLATFORM**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This application claims the benefit of priority to and is a continuation of U.S. patent application Ser. No. 18/512,382, filed Nov. 17, 2023, which claims the benefit of priority to and is a continuation of U.S. patent application Ser. No. 16/870,557, filed May 8, 2022, now U.S. Pat. No. 11,822,622, issued Nov. 21, 2023, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] This disclosure relates generally to an online document system, and more specifically to managing online document system feature accessibility.

BACKGROUND

[0003] An online document system provides access to features of the online document system to entities subscribed to use the features. For example, an engineering company is subscribed to use an electronic signature feature and may access it to electronically sign online documents. While the online document system may track information associated with how subscribed entities use features of the online document system (e.g., how frequently features are used and what features are used), the online document system may not have access to information detailing how the features might apply to non-subscribed entities.

[0004] This lack of information surrounding non-subscribed entities limits the accuracy of the online document system's feature management, for instance making it more difficult for the online document system to optimize feature recommendations provided to non-subscribed entities. Likewise, this lack of information may prevent potential customers from being reached. Accordingly, an online document system may benefit from having access to feature information associated with non-subscribed entities.

SUMMARY

[0005] The online document system described herein implements feature access control with entities associated with the online document system to improve feature management. In particular, the online document system uses feature access control to allow a first entity associated with the online document system (e.g., an entity subscribed to one or more features offered by the online document system, referred to herein as the "subscribed entity" or "parent entity") to sponsor a second entity that does not have access to the features (referred to herein as the "non-subscribed entity" or "child entity") so that the second entity can access features of the online document system. Access policies may be specified by the subscribed entity to set restrictions for the non-subscribed entity's access to sponsored features.

[0006] The online document system may obtain information on how entities use features of the online document system to improve feature management. For example, a feature recommendation engine of the online document system may train a machine learning model with the feature use information to improve the accuracy of feature recom-

mendations made by the online document system. The information may also increase a likelihood that non-subscribers convert to subscribers of the online document system, which may provide the online document system with more data about feature usage by the converted entities.

[0007] In one embodiment, the online document system provides access to an entity to use a sponsored feature. A first entity (e.g., an engineering company) that is subscribed to the online document system may have access to a first set of features within the system. The online document system receives, from the engineering company, a first request to sponsor a second entity (e.g., an electronics vendor) with access to a second set of features (such as a subset of the first set of features). The online document system may then fulfill the first request by providing the second entity with access to the second set of features. Later, the online document system receives a second request from the engineering company that identifies another feature to sponsor for the second entity. This identified feature may be one that the engineering company has access to but has not yet sponsored the electronics vendor to access. For instance, this identified feature may be included in the first set of features but not included in the second set of features.

[0008] Before fulfilling this second request, the online document system determines a first access policy and receives a second access policy from the engineering company. The first access policy is an access policy specified by the online document system that helps govern access to the identified feature. The second access policy is an access policy specified by the engineering company that helps govern access to the identified feature. In this example, both access policies govern whether the electronics vendor can access the identified feature. The policies include requirements that must be satisfied before the electronics vendor can access the identified feature. If the electronics vendor satisfies the requirements, the online document system provides the vendor access to the identified feature.

[0009] In another embodiment, the online document system trains and applies a machine learning model to identify one or more features to provide in a recommendation. The online document system accesses a set of feature training data. The feature training data can describe characteristics of entities associated with the online document system and historical activity associated with usage of features within the online document system by those entities. The online document system trains a machine learning model using the accessed set of feature training set. The machine learning model may be configured to identify at least one feature to recommend to a target entity (e.g., the engineering company). The recommendation may be made by the online document system based on characteristics of the target entity and activity associated with the usage of features of the online document system by entities similar to the target entity. The online document system identifies the target entity, for which it is providing a recommendation to, and a second entity associated with the target entity (for instance, through a professional relationship between the first entity and the second entity). The machine learning model is applied to characteristics of the target entity and the second entity to identify features to recommend. The recommended feature may be identified based in part by its popularity with other, similar entities. The online document system then provides the target entity with a recommendation that identifies this popular feature.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a block diagram of a system environment in which an online document system operates, in accordance with at least one embodiment.

[0011] FIG. 2 is a block diagram of the online document system of FIG. 1, in accordance with at least one embodiment.

[0012] FIG. 3 shows a training and application diagram of the feature recommendation engine of FIG. 2, in accordance with at least one embodiment.

[0013] FIG. 4 depicts a graphical user interface (GUI) for requesting access to features within the online document system, in accordance with at least one embodiment.

[0014] FIG. 5 depicts a notification provided in the GUI of FIG. 4 when requesting access to a feature, in accordance with at least one embodiment.

[0015] FIG. 6 illustrates the interactions that take place within the system environment of FIG. 1 when providing access to features within the online document system, in accordance with at least one embodiment.

[0016] FIG. 7 is a flowchart illustrating a process for providing an entity access to an identified feature, in accordance with at least one embodiment.

[0017] FIG. 8 is a flowchart illustrating a process for recommending a feature to an entity, in accordance with at least one embodiment.

[0018] The figures depict various embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

DETAILED DESCRIPTION

System Architecture

[0019] FIG. 1 is a block diagram of a system environment 100 in which the online document system 140 operates. The system environment 100 shown by FIG. 1 includes the online document system 140, a parent entity 120, child entities 121 and 122, and a third-party service 130. The system environment 100 may have alternative configurations than shown in FIG. 1, including for example different, fewer, or additional components. For example, an additional parent entity having a child entity may be communicatively coupled with the network 110 to the online document system 140. As another example, the online document system 140 may be communicatively coupled with a remote database storing data for the entities serviced (e.g., storing authorization information for using features of the online document system, documents, or entity profile information). As referred to herein, a “feature” is a service or functionality offered by the online document system 140. The features within the online document system 140 include, for example, features to communicate, edit, authenticate, and/or execute online documents. As referred to herein, an “online document” is an electronic document such as an invoice, contract, sales order, or agreement sent through a communications network (e.g., the network 110).

[0020] The system environment 100 described herein can be implemented within the online document system 140, a document execution system, or any type of digital transaction management platform. It should be noted that although

description may be limited in certain contexts to a particular environment, this is for the purposes of simplicity only, and in practice the principles described herein can apply more broadly to the context of any digital transaction management platform. Examples can include but are not limited to online signature systems, online document creation and management systems, collaborative document and workspace systems, online workflow management systems, multi-party communication and interaction platforms, social networking systems, marketplace and financial transaction management systems, or any suitable digital transaction management platform.

[0021] The network 110 may serve to communicatively couple the parent entity 120, the child entities 121 and 122, the third-party service 130, and the online document system 140. For example, the online document system 140 and the parent entity 120 are configured to communicate via the network 110. In some embodiments, the network 110 includes any combination of local area and/or wide area networks, using wired and/or wireless communication systems. The network 110 may use standard communications technologies and/or protocols. For example, the network 110 includes communication links using technologies such as Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), 3G, 4G, 5G, code division multiple access (CDMA), digital subscriber line (DSL), etc. Examples of networking protocols used for communicating via the network 110 include multiprotocol label switching (MPLS), transmission control protocol/Internet protocol (TCP/IP), hypertext transport protocol (HTTP), simple mail transfer protocol (SMTP), and file transfer protocol (FTP). Data exchanged over the network may be represented using any suitable format, such as hypertext markup language (HTML) or extensible markup language (XML). In some embodiments, all or some of the communication links of the network 110 may be encrypted using any suitable technique or techniques.

[0022] The parent entity 120 is an entity using the online document system 140 to communicate documents to or coordinate document activity with other entities (e.g., the child entities 121 and 122). Example entities include a retail business, an Internet retailer, a small business, a bureaucratic agency, an engineering company, an educational institution, a consulting firm, a healthcare provider, a bank, or any suitable combination thereof. As referred to herein, communication with an entity (e.g., a parent entity or a child entity) involves communication with a computing device of the entity, where the computing device is capable of receiving user input as well as transmitting and/or receiving data via the network 110. In some embodiments, the computing device of an entity (e.g., the parent entity 120 and the child entities 121 and 122) is a conventional computer system, such as a desktop or a laptop computer. Alternatively, the computing device may be a device having computer functionality, such as a personal digital assistant (PDA), a mobile telephone, a smartphone, or another suitable device. The computing device is configured to communicate with the online document system 140 via the network 110, for example using a native application executed by the computing device and provides functionality of the online document system 140, or through an application programming interface (API) running on a native operating system of the computing device, such as IOS® or ANDROID™.

[0023] The child entities **121** and **122** are entities that use the online document system **140** to communicate documents to or coordinate document activity with the parent entity **120**. The child entities **121** and **122** may not be directly subscribed to a feature of the online document system **140** used with the parent entity **120**. Features and subscriptions are described in further detail below. In some embodiments, the child entities **121** and **122** may have a professional relationship with the parent entity **120**. For example, the child entities **121** and **122** may be business partners with the parent entity **120**, may be students of an educational institution represented by the parent entity **120**, or may be banking customers of the parent entity **120**.

[0024] The third-party service **130** may be an entity independent of the online document system **140** and the entities **120**, **121**, and **122** that provides services to the online document system **140**. For example, the third-party service **130** verifies, on behalf of the online document system **140**, whether entities are authorized to access features of the online document system **140**.

[0025] The online document system **140** provides access to features of the online document system to entities associated with the online document system. In some embodiments, a feature is accessible to an entity upon satisfying an access policy. As referred to herein, an “access policy” is at least one requirement that an entity must satisfy to use a feature or features within the online document system **140**. The term “access policy” is used interchangeably herein with the term “policy.”

[0026] For example, the online document system provides the child entity **121** with access to a document authentication feature after the entity provides identity verification to satisfy the access policies established by the online document system and the parent entity **120**. In this example, the child entities **121** and **122** may not directly subscribe to the document authentication feature within the online document system **140** but can use the document authentication feature because of sponsorship from the parent entity **120**. Access policy creation and verification is described further in the description of FIG. 2. A process to sponsor an entity with access policies is described further in the descriptions of FIGS. 6 & 7. In some embodiments, the online document system **140** provides a GUI to the parent entity **120** to create a sponsorship. Examples of such a GUI are described further in the descriptions of FIGS. 4 and 5.

[0027] In some embodiments, the online document system **140** recommends features and access policies. For example, the online document system **140** uses machine learning to determine likely features and/or access policies based on characteristics of an entity. Characteristics of the entity may include profile information about the entity, activity history of the use of the online document system **140** and its features by the entity, or any suitable data related to the association of a feature or access policy to an entity. The training and application of a machine learning model for identifying a recommendation is described further in the description of FIG. 3.

[0028] The online document system **140** may recommend features for an entity to use itself or for an entity to use with other entities. A machine learning model may be applied to the parent entity **120**'s usage of features within the online document system **140** to determine a feature to recommend for the parent entity **120**. For example, the online document system **140** uses the parent entity **120**'s usage of an elec-

tronic signature service to identify a related document authentication feature to recommend.

[0029] Features can be recommended for an entity to use with other entities as well. The online document system may monitor the entities **120**, **121**, and **122** and receive data representing the interactions among the entities. A machine learning model may then be applied to the received data to identify a feature to recommend to one or more of the entities for both internal use and use with the other entities.

Online Document System

[0030] FIG. 2 is a block diagram of the online document system **140** of FIG. 1. The online document system **140** includes software modules such as a featurization module **210**, a security engine **215**, and a feature recommendation engine **220**. In some embodiments, the online document system **140** includes or accesses local databases such as a policy and feature database **200** and an entity profile database **205**. The online document system **140** may have alternative configurations than shown in FIG. 2, including different, fewer, or additional components. For example, at least one of the databases **200** or **205** may be stored remotely and may be accessible through the network **110**.

[0031] The policy and feature database **200** may include records of access policies available for authorizing an entity to use an available feature and records of the available features. Access policies may be characterized by one or more requirements. For example, an access policy is characterized by a requirement that a child entity use a sponsored feature with a parent entity only. Other examples of requirements include: one or more users of a child entity be whitelisted by the sponsoring parent entity (e.g., through a whitelisted domain address), a child entity use the sponsored feature within a predetermined period of time, the child entity use the sponsored feature no more than a maximum number of times, an identity of a user from the child entity be verified, or any suitable requirement adding security to and/or limiting the use of the sponsored feature.

[0032] Policies stored in the policy and feature database **200** may be accessible by the security engine **215** to verify an entity's authorization to use a feature, or by the featurization module **210** to generate a training set for a machine learning model configured to output recommended access policies. In some embodiments, access policies may be specified by both the online document system **140** and an entity (e.g., the parent entity **120**). For example, for the child entity **121** to use a sponsored document authentication feature, the security engine **215** determines whether the child entity **121** satisfies an access policy, determined by the online system **140**, that the child entity **121** is using the feature with the parent entity **120** and also satisfies an access policy, determined by the parent entity **120**, that a user of the child entity **121** provides identity credentials before accessing the feature.

[0033] The policy and feature database **200** may also store information associated with features available within the online document system **140**, such as an electronic signature service for executing documents, an identity authenticator for accessing and/or executing an online document, a setting configurator for configuring settings of the online document, a supplemental online document provider for attaching supplemental documents in an electronic envelope with an

online document, an online document resizer for adapting an aspect ratio of the online document to a device screen size, and the like.

[0034] Stored data in the entity profile database **205** includes entity characteristic information and activity history. As referred to herein, “characteristic information” represents characteristics of an entity that is not directly related to the online document system **140** and “activity history” represents activities performed by the entity involving the online document system **140**. Characteristic information may include entity types, a type and number of additional entities with which entities interact, applications used by entities to access online documents, whitelisted contacts of the entities for whom respective identities are pre-verified, geographic locations of the entities, a number of employees associated with each entity, a number and type of accounts associated with the online document of each entity, security measures associated with each entity, a trustworthiness associated with each entity, internet protocol (IP) addresses used by computing devices of the entities, hardware attributes of the computing devices of the entities, or any suitable characteristic of an entity that is not directly related to the online document system. For example, an engineering company may be characterized by its number of employees, location, trustworthiness based on a credit rating determined by a credit rating agency, and use of thousands of documents associated with account types such as “staff engineer.”

[0035] Activity history may include data representing the features used by the entities, a number of the entities using a feature, access policies used by the entities, a frequency with which an entity uses the features, times and durations during which the entity uses the features, monetary values associated with the usage of the features by the entities, sponsorships of the features by the entities on behalf of other entities, any suitable activity performed by the entity involving the online document system, or any combination thereof. For example, the activity history of the engineering company includes a representation of its use of a document execution feature, an average number of times in a day the document execution feature is used, and the fees paid by the engineering company to use the document execution feature.

[0036] In some embodiments, entity profile database **205** stores data associations involving at least one of an entity, a feature, and a policy. For example, the entity profile database **205** may have a look-up structure where access policies may be tagged with the entity and/or feature it is used for. Entities may specify sponsored features and a corresponding access policy through the online document system **140** (e.g., through GUI **400**) to be stored in the entity profile database **205**.

[0037] The featurization module **210** encodes data from the databases **200** and **205** for input into a machine learning model. For example, the online document system **140** may provide entity profile information and activity history to the featurization module **210**, which generates a mathematical representation of the data. The mathematical representation may be a multi-entry vector (a “feature vector”), with each entry representative of a characteristic of the training data. Each entry of the feature vector may be indicative of a particular data type, data magnitude, data value, or combination thereof. For example, a value of an element representing the number of employees of the parent entity **120** may be the literal number of employees or a number

representing the number of the parent entity **120**’s employees relative to other entities’ numbers of employees.

[0038] In some embodiments, the featurization module **210** applies weights to training sets or feature vectors input to a machine learning model. For example, the featurization module **210** may determine degrees of similarities between entities and apply weights based on the determined degrees, weighting a more similar entity greater than a less similar entity. In some embodiments, the featurization module **210** reduces the dimensions of a feature vector by replacing multiple vector elements with a weighted sum or average of the element values, which may reduce the number of computations performed by the feature recommendation engine **220**. The featurization module **210** may filter a subset of data from databases **200** and **205** to be generated into a feature vector. For example, the featurization module **210** encodes data that has been collected within the past year to improve the likelihood that outputs from the machine learning models reflect recent entity profiles and activity history.

[0039] The security engine **215** determines whether an entity satisfies an access policy associated with the use of a feature. The security engine **215** may access the entity profile database **205** to determine one or more access policies specified by the online document system **140** or the parent entity **120** for the child entity **121** to use a sponsored feature. For example, the security engine **215** determines that the parent entity **120**’s access policy requires that the child entity **121** provide password credentials to use a sponsored electronic signature service. Additionally, the security engine **215** may determine that the online document system **140**’s access policy requires that the child entity **121** only use the sponsored signature feature with the parent entity **120**. The security engine **215** may enforce the system **140**’s access policy by preventing users without an identification credential (e.g., an email address and/or password) associated with either the parent entity **120** or the child entity **121** from accessing documents modified using the sponsored signature feature. It is noted that an access policy may be further enforced even after a sponsored feature is accessed by an entity and may prevent the feature from being fully utilized until requirements of the access policy are satisfied. The process to verify and provide access to a feature is described further in the description of FIG. 6.

[0040] The feature recommendation engine **220** trains and applies a machine learning model to identify one or more features of the online document system **140** to recommend. The online document system **140** accesses data (e.g., in the database **205**) representing the usage of features of the online document system by various entities associated with the online document system. The featurization module **210** accesses and encodes this feature usage data stored in the entity profile database **205**. To train a machine learning model, the feature recommendation engine **220** accesses this encoded feature usage data to generate a training set. The machine learned model can be trained to identify correlations between the usage of particular features and characteristics of the entities that use the features.

[0041] The feature recommendation engine **220** may include multiple machine learning models, the models configured to accept different inputs to generate a feature to recommend. Machine learning models of the feature recommendation engine **220** may use various machine learning techniques such as linear support vector machine (linear SVM), boosting for other algorithms (e.g., AdaBoost), neu-

ral networks, logistic regression, naïve Bayes, memory-based learning, random forests, bagged trees, decision trees, boosted trees, boosted stumps, or any suitable supervised or unsupervised learning algorithm.

[0042] The identified feature output by the machine learning model may be based on similarities between characteristics of the entity receiving the recommendation and another entity associated with the identified feature. The feature recommendation engine 220 may determine these similarities based on comparisons of entity characteristics. Additionally, or alternatively, the feature recommendation engine 220 may determine similarities based on comparisons of activity history. For example, the feature recommendation engine 220 compares the parent entity 120's company size, the types of accounts associated with the online documents used, and the features it sponsors to similar information of other entities.

[0043] In some embodiments, the feature recommendation engine 220 may determine a degree of similarity quantifying the similarities between entities' respective characteristics and/or activity history. The feature recommendation engine 220 may use the determined degrees of similarities between entities to pre-process the feature vectors from the featurization module 210. For example, the feature recommendation engine 220 may determine degrees of similarity representative of the percentage of features sponsored by the parent entity 120 and by other entities. The feature recommendation engine 220 then uses the determined degrees of similarity to filter out or exclude feature vectors associated with entities that have a small degree of similarity from training sets. Additionally, or alternatively, the feature recommendation engine 220 may prioritize entities with larger degrees of similarity by using a training set that includes intentional duplicates of feature vectors associated with those entities.

[0044] While the online document system 140 is capable of providing recommendations for many features to an entity, the entity may not have access to sponsor every feature. In some embodiments, the identified feature output by the feature recommendation engine 220 is a feature that an entity has authorization to sponsor. The online document system 140 may limit the number of features that can be sponsored according to a subscription type. For example, the parent entity 120 may have access to a first set of features to sponsor by subscribing to the online document system 140 with a basic account. After subscribing with an advanced subscription account, the parent entity 120 may have access to the first set and an additional second set of features to. As described further in the description of FIG. 5, a particular subscription type may be one requirement for sponsoring a feature.

[0045] The identified feature output by the feature recommendation engine 220, in some embodiments, is identified based on features historically sponsored by entities subscribed to the online feature system 140. The feature recommendation engine 220 may provide a recommendation that includes a feature historically sponsored by an entity or currently sponsored by at least a threshold number of entities. For example, the feature recommendation engine 220 prevents features sponsored by less than 5 entities from being recommended and sorts the features that are recommended by the number of entities currently sponsoring them.

[0046] In some embodiments, the identified feature output by the feature recommendation engine 220 is a feature that, presently, a first entity has access to and a second entity does not. For example, a machine learning model of the feature recommendation engine 220 outputs a feature that the parent entity 120 has access to and the child entity 121 does not have access to.

[0047] The identified feature output by the feature recommendation engine 220, in some embodiments, is identified based on activity history involving a previously recommended feature. For example, the online document system 140 receives usage information indicating that the parent entity 120 is using a previously recommended electronic signature service with the child entity 121. The feature recommendation engine 220 may recommend a document authentication feature to the parent entity 120 based on its previous use of the previously recommended electronic signature service.

[0048] In some embodiments, the feature recommendation engine 220 uses activity history involving recommended features as feedback to re-train a machine learning model that outputs recommended features. For example, the feature recommendation engine 220 may strengthen an association between characteristics of the parent entity 120 and a recommended feature responsive to the parent entity 120 accepting the recommendation. Similarly, the feature recommendation engine 220 may weaken an association between characteristics of the parent entity 120 and the recommended feature responsive to the parent entity 120 not accepting the recommendation. The feature recommendation engine 220 may strengthen associations by re-training a machine learning model with updated training sets, generated by the online document system 140, that have an increased number of mappings of a feature vector characterizing an entity (e.g., the parent entity 120) and the recommended feature. Similarly, the feature recommendation engine 220 may weaken associations by re-training a machine learning model with updated training sets, generated by the online document system 140, that do not map or have a decreased number of mappings of a feature vector characterizing an entity (e.g., the parent entity 120) and the recommended feature.

[0049] The feature recommendation engine 220 may provide, to an entity, the identified feature output by a machine learning model in a recommendation. As a part of a recommendation generation process, the feature recommendation engine 220 may identify which entity to provide the recommendation to, when to provide the recommendation, and/or a frequency with which to provide recommendations to an entity.

[0050] In some embodiments, the feature recommendation engine 220 identifies one or more entities to provide the recommendation to. For example, the feature recommendation engine 220 identifies a parent-child entity pair to recommend the identified feature to at least one entity of the pair. This identification may be performed by accessing the entity profile database 210 to determine an entity that is sponsoring a feature and lookup an entity for whom the feature is sponsored for. The feature recommendation engine 220 may identify entities to provide a recommended feature based on their previous interactions with each other (e.g., records of entity business interactions and/or interactions involving the online document system 140 stored within the entity profile database 205). The feature recommendation

engine 220 may target a recommendation for the parent entity 210 and identify the child entity 212 as an additional or alternative entity to provide a recommendation for.

[0051] In some embodiments, the feature recommendation engine 220 can identify entities to provide a recommended feature to upon receiving an indication that an entity is requesting a feature to be recommended for use with another entity. Using this indication, the feature recommendation engine 220 may identify both the requesting entity and the entity for whom the request is made. The indication may be received through a GUI provided for display by the online document system 140. For example, a selection of an entity to provide with access to a feature to is an indication of a request communicated to the online document system 140. In some embodiments, information describing restrictions and/or requirements to accessing recommended features are presented in the GUI. An example GUI is described in the description of FIGS. 4 and 5.

[0052] In some embodiments, the online document system 140 determines an entity and a time at which to provide a recommendation. An entity's activity history and/or usage of recommended features may be used for this determination. For example, the online document system 140 determines to recommend a document auto-population feature to the parent entity 120 at a time when the parent entity 120 is accessing a document with many text fields. In another example, the online document system 140 may determine a time to provide a recommendation independent of the target entity to which the recommendation is to be provided. For example, the online document system 140 may periodically monitor feature usage across all entities to determine the popularity of features. In turn, the online document system 140 determines the time that a particular feature is popular to recommend a related feature to all entities.

[0053] In some embodiments, the feature recommendation engine 220 uses activity history involving recommended features as feedback to modify how it provides recommendations. The feature recommendation engine 220 may determine a quantity and/or frequency at which to provide recommendations based on a frequency with which an entity uses recommended features. For example, responsive to determining that the frequency at which the entity uses recommended features is lower than a threshold frequency, the feature recommendation engine 220 may stop providing recommendations for a threshold period of time.

[0054] In some embodiments, the feature recommendation engine 220 may modify how it provides recommendations based on the monetary value with which an entity is spending on features. For example, if the parent entity 120 is spending a monetary value above a threshold value on features of the online document system 140, the feature recommendation engine 220 may increase the frequency with which it provides feature recommendations to the parent entity 120.

[0055] Although not depicted in FIG. 2, the online document system 140 may include an access policy recommendation engine. This access policy recommendation engine may identify policies for recommendation or for default use by the online document system 140 and store those identified policies within the policy and feature database 200. While an entity may request a recommended access policy, the output of the access policy recommendation engine may be enforced by default by the online document system 140.

A default access policy may be enforced if an entity has not yet specified an access policy.

[0056] The access policy recommendation engine may include a machine learning model configured to identify and/or reference an access policy from the policy and feature database 200 to output as a recommendation to an entity. Similar to a machine learning model used in the feature recommendation engine 220, a machine learning model of the access policy recommendation engine may be trained based on historical entity characteristics information and applied to subsequently received entity characteristics information to map the data to an access policy. For example, the access policy recommendation engine is trained using data sets including historical feature usage data and/or data representing characteristics of the entities generating the usage data. The access policy recommendation engine may then receive a feature vector representative of present entity profile information (e.g., currently monitored feature usage data and/or updated characteristics of entities generating the usage data) as input and may map an access policy to entries of the feature vector. The access policy recommendation engine determines, based on the output of the machine learning model, an access policy associated with the entity profile data input to recommend.

[0057] In some embodiments, the access policy recommendation engine determines, based on an entity's usage history with features of the system 140, an adjustment to an existing access policy of the entity. The access policy recommendation engine may then provide the adjustment to an existing access policy. For example, the access policy recommendation engine may determine that the child entity 121 is exceeding a threshold request frequency for requesting to use a feature sponsored by the parent entity 120 after a predetermined period of time for granted access has expired. In response, the access policy recommendation engine may use the data representative of the request frequency to determine an access policy with a longer time before granted access to the feature expires (e.g., an additional day).

[0058] The access policy recommendation engine may identify policies based on a subscription. The online document system 140 may provide access to premium access policies for entities that have a premium subscription. For example, the parent entity 120 may pay a subscription fee in exchange for a premium access policy that requires an identity be authenticated using a reputed third-party security service.

[0059] FIG. 3 shows a training and application diagram 300 of the feature recommendation engine 220. The online document system 140 receives training data 310 from entities associated with the online document system 140. To avoid biased outputs, the training data 310 used to recommend features to the entities 120, 121, and 122 may not directly be provided by those entities themselves. The feature recommendation engine 220 determines recommended features 320 based on the received data and provides them to the parent entity 120. The training and application diagram 300 may have alternative configurations than shown in FIG. 3, including different, fewer, or additional components. For example, the recommended features 320 may be provided directly to the child entities 121 and 122.

[0060] The parent entity 120 is subscribed to features 330 of the online document system 140 and is associated with customized access policies 331 and 332. The parent entity

120 may communicate electronic envelopes **340** and **350** to the child entities **121** and **122**, respectively. These electronic envelopes **340** and **350** can contain online documents associated with the online document system **140** (e.g., edited using or maintained by the online document system **140**). The child entities **121** and **122** are sponsored by the parent entity **120** to use features **360** and **370**, respectively.

[0061] The online document system **140** may store the training data **310** in the entity profile database **205**. Although labeled as “training data **310**,” the data received has not necessarily been prepared or processed for the purposes of training a machine learning model of the online document system **140**. Rather, the label “training data **310**” includes data in both a pre-processed and processed form that is relied upon to train a machine learning model of the online document system **140**. The featurization module **210** may access the training data **310** from the entity profile database **205** to encode into feature vectors for training a machine learning model used by the feature recommendation engine **220**.

[0062] A usage history **380** of the entities **120**, **121**, and **122** may be received by the online document system **140** after the machine learning model has been trained. The usage history **380** may include the use of the sponsored features **360** and **370** and the use of the customized access policies **331** and **332**. Although not depicted, additional entity characteristic information such as the properties of the online documents communicated within the electronic envelopes **340** and **350** may be communicated to the online document system **140**.

[0063] The usage history **380** may be stored into the entity profile database **205** and encoded by the featurization module **210**. The feature recommendation engine **220** may then input the encoded data into the trained machine learning model of the feature recommendation engine **220** to output the recommended features **320** to the parent entity **120**. Although not depicted, an access policy recommendation engine may receive the encoded data for input into a machine learning model trained to output an access policy for recommendation. The recommended access policy may be provided by the online document system **140** to the parent entity **120**.

Feature Management Interface

[0064] FIGS. 4-5 depict a GUI **400** for managing access to features within the online document system **140** of FIG. 1. The GUI **400** may be presented to an entity that is requesting to sponsor a feature for another entity. For example, the parent entity **120** may use the GUI **400** to edit which features it is sponsoring for the child entities **121** and **122**. In alternative configurations, different and/or additional components may be included in the GUI **400**. For example, filtering toggles may be included to sort and/or limit entities, features, or access policies displayed in GUI **400**.

[0065] The GUI **400** includes an entity selection field **410**, a feature list **420**, an access policy editing box **430**, and a request sponsorship button **440**. An entity subscribed to features of the online document system **140** may specify one or more entities for which to sponsor selected features using specified policies. For example, the parent entity **120** may specify the child entity **121** and/or the child entity **122** in the entity selection field **410**. The parent entity **120** may then select, from the feature list **420**, features to sponsor for the specified entity or entities within the entity selection field

410. The access policy editing box **430** allows the parent entity to create or edit access policies for the specified entity or entities to access the selected features. The GUI **400** may automatically populate with at least one of a recommended entity, feature, or access policy. The parent entity **120** may select the request sponsorship button **440** to transmit the request for sponsorship as defined by the selections made within the GUI **400**.

[0066] A feature **421** and a premium feature **422** have been selected in the feature list **420**. The selection is indicated by the presence of shading within the corresponding text boxes. As referred to herein, a “premium feature” is a feature within the online document system **140** with more access restrictions than other, non-premium features. Requirements to use premium features may include restrictions to particular entity characteristics (e.g., limited to entities located in North America), a minimum duration of subscription with the online document system **140**, a premium subscription fee, a promotional code, existing access to one or more specific features, any suitable requirement for using a feature, or any combination thereof. Information regarding access restrictions or requirements associated with premium features may be provided through GUI **400**, as described in the description of FIG. 5. The online document system **140** may list available features (e.g., as stored in the policy and feature database **200**) in the feature list **420**. The online document system **140** may visually distinguish premium features listed in the feature list **420**. As depicted in FIG. 4, premium features are visually distinguished using an icon (e.g., a star) next to each premium feature listing. Visual distinguishers are not limited to icons. Differing colors, font sizes, fonts, animations, or any combination thereof may be used to visually distinguish premium features in the GUI **400**.

[0067] The online document system **140** may automatically populate at least one of the entity selection field **410**, the feature list **420**, or the access policy editing box **430**. The online document system **140** may identify an entity to sponsor a feature for based on the interactions between the parent entity **120** and other entities. The online document system **140** may then populate the entity selection field **410** with this identified entity. The contents of the feature list **420** may be populated based on outputs of the feature recommendation engine **220**.

[0068] The online document system **140** provides the access policy editing box **430** for the creation or edit of access policies associated with the selected entity in the entity selection field **410** and selected features in the feature list **420**. The access policy editing box **430** is depicted with an access policy **431** configured. The access policy **431** is characterized by a password requirement, where the password may be provided through either one of two verification methods (e.g., a user specified password or a temporary password generated by the online document system and provided to a secured user account). The parent entity **120** may select the request sponsorship button **440** to transmit the request for sponsorship as defined by the selections made within the GUI **400**.

[0069] FIG. 5 depicts a notification provided in the GUI **400** when requesting access to a feature. The GUI **400** of FIG. 4 is depicted with a mouse icon controlled by the parent entity **120** over the premium feature **422** within the feature list **420**. Responsive to receiving a selection of the premium feature **422** and/or upon detecting that the mouse icon is

hovering over the premium feature **422**, the online document system **140** may provide a popup **500** for display. The popup **500** includes information describing restrictions and/or requirements for sponsoring the premium feature **422**. For example, the parent entity **120** may be required to subscribe with a premium subscription fee before having access to sponsor the premium feature **422**.

Processes for Sponsoring Features Within the Online Document System

[0070] FIG. 6 illustrates an embodiment of interactions **600** that take place within the system environment **100** of FIG. 1 when providing access to features within the online document system **140**. The parent entity **120** and the child entity **121** interact with the online document system **140** such that the child entity **121** can access features sponsored by the parent entity **120**. When the child entity **121** requests to access sponsored features, the online document system **140** may verify that the child entity **121** satisfies access policies to fulfill the request. The interactions of FIG. 6 may be facilitated by communication over the network **110**.

[0071] In some embodiments, the interactions illustrated in FIG. 6 involves additional, fewer, or different functions or entities for performing the functions. For example, the third-party service system **130** may interact with the online document system **140** to verify that the child entity **121** satisfies the appropriate access policies. In another example, although the access policy determined by the online document system **140** is depicted as being determined prior to receiving the access policy provided **625** by the parent entity **120**, the online document system **140** may receive a policy specified by the parent entity **120** prior to determining its own access policy. In other alternative embodiments, only one of the first or second access policies is determined or no access policy is determined for the child entity **121** to access the additional features.

[0072] The parent entity **120** initiates the interactions by requesting **605** to sponsor the child entity **121** to access a set of features. The online document system **140** receives the request and in response, provides **610** the child entity **121** with access to the set of features. For example, the parent entity **120** may request to sponsor electronic signature and document authentication features for the child entity **121**. The online document system **140** receives this request and provides the child entity **121** access to the electronic signature and authentication features. Although not depicted between the requesting **605** and the providing **610**, one or more access policies may be specified by the parent entity **120** and/or the online document system **140** to enforce when the child entity **121** accesses the features provided **610**.

[0073] The parent entity **120** later requests **615** to sponsor the child entity **121** to access additional features (such as one or more premium features). The online document system may determine **620** a first access policy and receive a second access policy provided **625** by the parent entity **120**. For example, the parent entity **120** requests **615** to sponsor an additional supplemental document feature for the child entity **121** to access. The online document system **140** may determine **620** a first access policy with a requirement that the child entity **121** only use the sponsored supplemental document feature with the parent entity **120**. The parent entity **120** provides **625** a second access policy to the online document system **140** that requires that the child entity **121**

use the sponsored supplemental document feature prior to a predetermined expiry time (e.g., a month from the request **615**).

[0074] After the first and second access policies are established, the child entity **121** attempts **630** to access the additional features. The online document system **140** verifies **635** that the child entity **121** satisfies the access policies. After verifying **635** that the child entity **121** has satisfied access policies, the online document system **140** provides **640** access to the additional features. Finally, the child entity **121** accesses **645** the additional features. For example, the child entity **121** may attempt to attach supplemental documents to an electronic envelope, such as an email, using the sponsored feature. The attempt **630** may be in the form of a request from the child entity **121** and received by the online document system **140**. Before fulfilling the request, the online document system **140** determines whether the expiry date specified in the second access policy by the parent entity **120** has passed and whether the child entity **121** is sending the electronic envelope to the parent entity **120**. When the online document system **140** verifies that the child entity **121** meets the requirements of the first and second access policies, the online document system **140** provides **640** the child entity **121** access to the sponsored feature. The child entity **121** may then receive a confirmation that the supplemental documents were attached to the electronic envelope (e.g., the online document system **140** provides the confirmation).

[0075] Although the verifying **635** is depicted as occurring before providing **640** access to the additional features, verifying **635** may, depending on the access policy, be finalized after the child entity **121** accesses the feature. For example, the online document system **140** may allow the child entity **121** to attach supplemental documents using the sponsored feature before determining whether or not the child entity **121** has used the sponsored feature with the parent entity **120**.

[0076] FIG. 7 is a flowchart illustrating a process **700** for providing an entity access to an identified feature of the online document system **140**. In some embodiments, the online document system **140** performs operations of the process **700** in parallel or in different orders, or performs different steps. For example, although not depicted, the process **700** may include determining an access policy for providing access to the set of features before providing **702** access to those features.

[0077] The online document system **140** determines **701** whether it has received a request from a first entity to sponsor a second entity. For example, the online document system **140** may determine whether it has received a request from the parent entity **120** to sponsor the child entity **121**. If the online document system **140** determines it has not received the request, the process **700** will not proceed to providing **702** access to a set of features and may repeat the determination **701** of whether it has received a request to sponsor an entity. If the online document system **140** determines it has received the request to sponsor a second entity, the process **700** continues. For example, the online document system **140** may receive a user selection of the request sponsorship button **440** of the GUI **400**, indicating that an entity is requesting to sponsor a set of features.

[0078] The online document system **140** provides **702** access to a set of features within the online document system **140**. For example, the online document system **140** provides

the child entity **121** access to a set of features that the parent entity **120** has requested to sponsor. The set of features may be selected from features that the parent entity **120** has access to. For example, the parent entity **120**, through its subscription to the services of the online document system **140**, has access to electronic signature and document authentication features and can request that the online document system **140** provide access to those features for the child entity **121** via sponsorship.

[0079] The online document system **140** receives **703** a second request to sponsor an identified feature for the second entity. For example, the parent entity **120** transmits another request to the online document system **140** to sponsor a feature it has not yet sponsored for the child entity **121**. The parent entity **120** may use the GUI **400** to request the sponsorship of a premium document auto-population feature for which it meets requirements to sponsor. For example, a popup similar to the popup **500** may not appear to notify the parent entity **120** as it attempts to sponsor the premium document auto-population feature because it meets the requirement of paying a premium subscription fee.

[0080] The online document system **140** accesses **704** first and second access policies including requirements that must be satisfied before the second entity can access the identified feature. The online document system **140** may determine the first access policy for the parent entity **120**'s request to sponsor the document auto-population feature. For example, the online document system **140** may determine an access policy including the requirement that the child entity **121** only use the document auto-population feature with the entity that sponsored the feature, the parent entity **120**. This determination can be made using the access policy recommendation engine as described herein. In other embodiments, the first access policy may be a default access policy used for all features or this specific feature. The parent entity **120** may determine the second access policy. For example, the parent entity **120** may provide an access policy to the online document system **140** using the GUI **400**. The second access policy may specify a requirement that the child entity **121** use the document auto-population feature within one month of the date on which the sponsorship was requested.

[0081] The online document system **140** determines **705** whether the second entity satisfies both the first and second requirements. For example, the online document system **140** determines whether the child entity **121** meets the first access policy requiring that the document auto-population feature only be used with the parent entity **120** and whether the child entity **121** meets the second access policy requiring that the document auto-population feature be used before a predetermined expiry date. If the child entity **121** does not satisfy the requirements of both access policies, the process **700** may not continue, and if the child entity **121** satisfies the requirements of both access policies, the process **700** continues.

[0082] The online document system **140** provides **706** the second entity access to the identified feature. After the child entity **121** has satisfied both access policies, it may access the document auto-population feature. The online document system **140** may provide a confirmation that access was granted. For example, the online document system **140** may begin automatically populating fields of an online document in accordance with the functionality of the document auto-population feature. As another example, the online document system **140** may provide a message (e.g., an email) to

the child entity **121** indicating that the online document with automatically populated fields was successfully sent to the parent entity **120**.

[0083] FIG. **8** is a flowchart illustrating a process **800** for recommending a feature of the online document system **140** to an entity. In some embodiments, the online document system **140** performs operations of the process **800** in parallel or in different orders, or may perform different steps. For example, although not depicted, the process **800** may include receiving data describing updated characteristics of entities and subsequent activity between entities before applying **804** the machine learning model to identify one or more features to recommend.

[0084] The online document system **140** accesses **801** a set of feature training data describing characteristics of entities and historical activity associated with usage of features by the entities. In some embodiments, the online document system maintains data representative of entity characteristics and historical activity within the entity profile database **205**.

[0085] The online document system **140** trains **802** a machine learning model using the accessed **801** feature training data. The data maintained in entity profile database **205** may be accessed for training **802** a machine learning model (e.g., of the feature recommendation engine **220**). Although not depicted in the process **800**, the featurization module **210** may be used to encode the data into suitable and/or standardized data prior to training **802** the machine learning model.

[0086] The online document system **140** identifies **803** a first entity and a second entity associated with the first entity. In some embodiments, the online document system **140** identifies entities using activity history between entities. For example, the online document system **140** may identify the parent entity **120** and subsequently identify the child entity **121** because records within the entity profile database **205** indicate that the parent entity **120** has sent an online document to the child entity **121**. In some embodiments, the online document system **140** receives an indication from the parent entity **120** that it intends to sponsor features for the child entity **121**.

[0087] The online document system **140** applies **804** the machine learning model to characteristics of the first entity or second entity to identify one or more features to recommend. For example, the online document system **140** applies the trained machine learning model from the training **802** to entity characteristics of the parent entity **120** and/or the child entity **121** to identify a premium document auto-population feature to recommend. In this example, the online document system **140** may monitor the number of fields within online documents communicated between the parent entity **120** and the child entity **121** to determine that at least a threshold number of documents communicated have fields that may be auto-populated using the premium document auto-population feature. The machine learning model of the feature recommendation engine **220** may be trained to map the document auto-population feature to entity characteristics representative of these monitored interactions.

[0088] The online document system **140** provides **805** the identified one or more features in a recommendation to the first entity. In some embodiments, the online document system **140** provides the recommendation through a GUI (e.g., the GUI **400**).

ADDITIONAL CONSIDERATIONS

[0089] The foregoing description of the embodiments has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the patent rights to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

[0090] Some portions of this description describe the embodiments in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, micro-code, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

[0091] Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

[0092] Embodiments may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

[0093] Embodiments may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

[0094] Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the patent rights. It is therefore intended that the scope of the patent rights be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the patent rights, which is set forth in the following claims.

What is claimed is:

1. A computer-implemented method, comprising:
 - generating, using at least one processor, one or more feature access policies for providing access to one or more features;
 - receiving, using the at least one processor, a request to access at least one feature in the one or more features;
 - selecting, based on one or more entity characteristics associated with the request, at least one feature access policy in the one or more feature access policies; and
 - providing, using the at least one processor, access to the at least one feature based on the at least one feature access policy.
2. The method of claim 1, wherein the at least one feature includes at least one of: a service, a functionality, an application, or any combination thereof.
3. The method of claim 2, wherein the at least one feature is associated with at least one of: an online document system, one or more electronic documents, an entity profile information associated with one or more entities, or any combination thereof.
4. The method of claim 3, wherein the one or more entity characteristics are associated with an entity in a plurality of entities.
5. The method of claim 4, wherein the one or more entity characteristics include at least one of: an entity type associated with the entity, a type of another entity with which the entity interacts or interacted, a number of additional entities with which the entity interacts or interacted, one or more applications used by the entity to access one or more online documents, one or more properties of one or more online documents communicated to and/or from the entity, one or more contacts of the entity, one or more geographic locations of the entity, a number of employees associated with the entity, a number and type of accounts associated with one or more online documents of the entity, one or more security measures associated with the entity, a trustworthiness associated with the entity, one or more internet protocol (IP) addresses used by computing devices of the entity, one or more hardware attributes of one or more computing devices of the entity, one or more characteristics of the entity that is not directly related to the online document system, or any combination thereof.
6. The method of claim 4, wherein the at least one feature includes at least one of: a feature available to the entity for sponsoring on behalf of another entity, a feature sponsored by the plurality of entities, a feature that another entity is authorized to use and the entity is not yet authorized to use, a feature sponsored on behalf of at least a threshold number of entities in the plurality of entities, or any combination thereof.
7. The method of claim 4, wherein the providing includes granting access to the entity to the at least one feature upon determining satisfaction of the at least one feature access policy.
8. The method of claim 7, further comprising providing access to the entity to at least another feature in the one or more features upon determining satisfaction of the at least one feature access policy.
9. The method of claim 2, wherein the at least one feature includes at least one of: generation of an electronic document, modification of an electronic document, authentication of an electronic document, communication of an electronic document, execution of an electronic document, or any combination thereof.

10. The method of claim **1**, wherein the receiving includes applying a machine learning model to the one or more entity characteristics to determine the at least one feature.

11. The method of claim **10**, wherein the selecting includes mapping, based on determining the at least one feature, the one or more entity characteristics to the at least one feature access policy.

12. A system, comprising:

at least one processor; and

at least one non-transitory storage media storing instructions, that when executed by the at least one processor, cause the at least one processor to:

receive a request to access at least one feature in one or more features, the at least one feature includes at least one of: a service, a functionality, or any combination thereof;

determine, based on the request, one or more entity characteristics;

map, based on the at least one feature, the one or more entity characteristics to at least one feature access policy for accessing the at least one feature; and

provide access to the at least one feature based on the at least one feature access policy.

13. The system of claim **12**, wherein the at least one feature is associated with at least one of: an online document system, one or more electronic documents, an entity profile information associated with one or more entities, or any combination thereof.

14. The system of claim **13**, wherein the one or more entity characteristics are associated with an entity in a plurality of entities.

15. The system of claim **14**, wherein the one or more entity characteristics include at least one of: an entity type associated with the entity, a type of another entity with which the entity interacts or interacted, a number of additional entities with which the entity interacts or interacted, one or more applications used by the entity to access one or more online documents, one or more properties of one or more online documents communicated to and/or from the entity, one or more contacts of the entity, one or more geographic locations of the entity, a number of employees associated with the entity, a number and type of accounts associated with one or more online documents of the entity, one or more security measures associated with the entity, a

trustworthiness associated with the entity, one or more internet protocol (IP) addresses used by computing devices of the entity, one or more hardware attributes of one or more computing devices of the entity, one or more characteristics of the entity that is not directly related to the online document system, or any combination thereof.

16. The system of claim **14**, wherein the at least one feature includes at least one of: a feature available to the entity for sponsoring on behalf of another entity, a feature sponsored by the plurality of entities, a feature that another entity is authorized to use and the entity is not yet authorized to use, a feature sponsored on behalf of at least a threshold number of entities in the plurality of entities, or any combination thereof.

17. The system of claim **14**, wherein providing access includes granting access to the entity to the at least one feature upon determining satisfaction of the at least one feature access policy.

18. The system of claim **17**, wherein providing access including providing access to the entity to at least another feature in the one or more features upon determining satisfaction of the at least one feature access policy.

19. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by at least one processor, cause the at least one processor to:

receive a request to access one or more features of an online document system;

determine one or more entity characteristics based on the request;

apply a machine learning model to the one or more entity characteristics to determine at least one feature in the one or more features;

select at least one feature access policy in one or more feature access policies for accessing the at least one feature by mapping the one or more entity characteristics to the at least one feature access policy; and

provide access to the at least one feature based on the at least one feature access policy.

20. The non-transitory computer-readable storage medium of claim **19**, wherein the at least one feature includes at least one of: a service, a functionality, or any combination thereof.

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